

Lab 9a: Charge to Mass Ratio of an Electron

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Data Collection & Analysis

Table of Data

k	Voltage ΔV_k	Current I_k	Radius ¹ r_k
1	400 V	1.8 A	0.090 m
2	270 V	1.4 A	0.090 m
3	220 V	1.6 A	0.070 m
4	200 V	1.8 A	0.060 m
5	250 V	1.6 A	0.080 m

For each trial k , the magnetic field strength B_k was approximated using:

$$B_k = \frac{\alpha N \mu_0 I_k}{R},$$

where $N = 130$ is the number of turns in the solenoid, I_k is the current for trial k , $R = 0.273$ m is the measured radius of the *solenoid*, and $\alpha = 0.7155$ is a unitless constant for our specific apparatus.

The charge-to-mass ratio q/m for each trial was calculated using the formula:

$$\frac{q}{m} = \frac{2\Delta V_k}{B_k^2 r_k^2},$$

where $(\Delta V_k, r_k)$ are the accelerating voltage and electron path radius for trial k .

The formula above is derived in the conclusion.

The calculated B_k and q/m values for each trial are as follows:

k	Magnetic Field Strength B_k	Charge-to-Mass Ratio q/m
1	0.77 mT	$1.663 \cdot 10^{11}$ C/kg
2	0.60 mT	$1.876 \cdot 10^{11}$ C/kg
3	0.69 mT	$1.948 \cdot 10^{11}$ C/kg
4	0.77 mT	$1.861 \cdot 10^{11}$ C/kg
5	0.69 mT	$1.658 \cdot 10^{11}$ C/kg

¹This is the radius of the electron's circular path, NOT the radius of the solenoid coils!

Comparison with Accepted Value

Recall that an electron has a charge of $-1.602 \cdot 10^{-19}$ C and a mass of $9.109 \cdot 10^{-31}$ kg.

Thus, the theoretical charge-to-mass ratio for an electron is:

$$\frac{q}{m} = \frac{1.602 \cdot 10^{-19} \text{ C}}{9.109 \cdot 10^{-31} \text{ kg}} = 1.759 \cdot 10^{11} \text{ C/kg}.$$

The percent error for each trial is:

<i>k</i>	Experimental q/m	Percent Error
1	$1.663 \cdot 10^{11} \text{ C/kg}$	5.45%
2	$1.876 \cdot 10^{11} \text{ C/kg}$	6.67%
3	$1.948 \cdot 10^{11} \text{ C/kg}$	10.78%
4	$1.861 \cdot 10^{11} \text{ C/kg}$	5.84%
5	$1.658 \cdot 10^{11} \text{ C/kg}$	5.72%

Conclusion

1. What are the possible sources of error in the e/m experiment?

- Human error in measuring r_k , the radius of the electron's path, which was difficult to measure accurately due to the "fuzziness" of the electron trail and parallax error.
- The solenoid is not ideal, so the magnetic field is not perfectly uniform and does not perfectly follow the formula $B = \alpha N \mu_0 I / R$.
- We are not in a perfect vacuum, so magnetic permeability is not exactly μ_0 .

2. Will the Earth's magnetic field be a factor in e/m experiment? Explain.

Although it would have an effect on the net magnetic field, the effect would be so miniscule compared to the magnetic field generated by the solenoid that it can be safely ignored.

Earth's magnetic field is around 25 μT , while the magnetic field generated by our solenoid is around 0.7 mT = 700 μT . Thus, the Earth's magnetic field is about 25 times weaker than the solenoid's magnetic field, so its effect on the experiment is negligible.

3. Using your values for accelerating voltage and magnetic field, determine the theoretical radius for a proton in the apparatus.

Take the equation for q/m and rearrange it to solve for r :

$$\begin{aligned}\frac{q}{m} &= \frac{2\Delta V}{B^2 r^2} \\ r^2 &= \frac{2\Delta V}{B^2 (q/m)} \\ r &= \sqrt{\frac{2\Delta V}{B^2 (q/m)}} = \sqrt{\frac{2\Delta V m}{B^2 q}}.\end{aligned}$$

We know that a proton has the same magnitude of charge as an electron, $q = 1.602 \cdot 10^{-19} \text{ C}$, and a mass of $m = 1.673 \cdot 10^{-27} \text{ kg}$.

Applying this formula to the data from our 5 trials:

k	ΔV_k	B_k	Theoretical Radius r
1	400 V	0.77 mT	0.375 m
2	270 V	0.60 mT	0.398 m
3	220 V	0.69 mT	0.316 m
4	200 V	0.77 mT	0.265 m
5	250 V	0.69 mT	0.333 m

4. Derive the q/m equation.

Let a particle with charge q and mass m be accelerated through a potential difference ΔV . By energy conservation, the kinetic energy gained by the particle is equal to the electric potential energy lost:

$$q\Delta V = \frac{1}{2}mv^2.$$

This means that the speed v of the particle after being accelerated is:

$$v = \sqrt{\frac{2q\Delta V}{m}}.$$

Now, if the particle enters a region with a uniform magnetic field B that is perpendicular to its velocity, it will experience a magnetic force $F = qvB$ that causes it to move in a circular path. That is, F is also the centripetal force which keeps the particle moving in a circle of radius r :

$$qvB = \frac{mv^2}{r}$$

$$qB = \frac{mv}{r}.$$

Substituting the expression for v from the energy equation into the magnetic force equation, we have:

$$qB = \frac{m}{r} \sqrt{\frac{2q\Delta V}{m}}$$

$$q^2 B^2 = \frac{m^2}{r^2} \frac{2q\Delta V}{m}$$

$$q^2 B^2 = \frac{2mq\Delta V}{r^2}$$

$$qB^2 r^2 = 2m\Delta V$$

$$\frac{q}{m} = \frac{2\Delta V}{B^2 r^2}.$$

5. Pre-lab: Why is it important to measure charge-to-mass ratio (e/m)?

Measuring the charge-to-mass ratio e/m of the electron is important because the charge e and the mass m are difficult to measure independently. The ratio e/m can be used to determine the motion of charged particles in electric and magnetic fields without needing to know that individual values of e and m .